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## NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

### Fifth Semester B.E. (Civil) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

#### 20CV505 - ENVIRONMENTAL ENGINEERING - I

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

|                   |                                                                                                                                                                                    | Marks | BT* | CO* | PO*   |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-------|
| <b>Unit – I</b>   |                                                                                                                                                                                    |       |     |     |       |
| 1.                | a) Discuss the importance of protected water supply.                                                                                                                               | 10    | L*1 | 1   | 1,2,3 |
|                   | b) Give the drinking water standards for the following parameter.<br>Recall their effects when they exceed their limits: i) Chlorides<br>ii) iron iii) Cadmium iv) Lead v) Mercury | 10    | L2  | 2   | 1,2,3 |
| 2.                | a) Elaborate factors influencing for intake structures site selection.                                                                                                             | 10    | L2  | 2   | 1,2,3 |
|                   | b) How human activities contribute to environmental pollution?                                                                                                                     | 10    | L2  | 1   | 1,2   |
| 3.                | a) Discuss the factors affecting Per capita Demand.                                                                                                                                | 10    | L3  | 1   | 1,2   |
|                   | b) Deduce Flow Chart of Water Treatment Plant                                                                                                                                      | 10    | L3  | 2   | 1,2,3 |
| <b>Unit – II</b>  |                                                                                                                                                                                    |       |     |     |       |
| 4.                | a) Explain the importance of Jar Test with the aid of neat sketch.                                                                                                                 | 10    | L2  | 3   | 1,2,3 |
|                   | b) Differentiate between lime soda and zeolite process.                                                                                                                            | 10    | L2  | 4   | 1,2   |
| 5.                | a) Draw plain sedimentation tank sketch and mention the advantages of it.                                                                                                          | 10    | L3  | 3   | 1,2,3 |
|                   | b) Enumerate Fluoridation and De-fluoridation process.                                                                                                                             | 10    | L2  | 4   | 1,2   |
| 6.                | a) Discuss about points of chlorination and Break point chlorination.                                                                                                              | 10    | L2  | 4   | 1,2   |
|                   | b) Explain four troubles in operating rapid sand filters.                                                                                                                          | 10    | L2  | 3   | 1,2,3 |
| <b>Unit – III</b> |                                                                                                                                                                                    |       |     |     |       |
| 7.                | a) What are the criteria for a good distribution system?                                                                                                                           | 10    | L1  | 5   | 1,2,3 |
|                   | b) Explain the importance of fire hydrants in water supply.                                                                                                                        | 10    | L1  | 5   | 1,2,3 |
| 8.                | a) Explain layouts of distribution network with advantages and disadvantages.                                                                                                      | 10    | L2  | 5   | 1,2,3 |
|                   | b) With a help of neat sketch demonstrate methods of supply of water?                                                                                                              | 10    | L2  | 5   | 1,2,3 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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